

Solids, Circles & Triangles



SYLLABUS 3.1 3.2 3.3 3.4

About a thousand years ago, a scholar climbed a hill in what is now Pakistan and measured the size of the Earth.

Al-Bīrūnī knew the older method. Eratosthenes had compared shadows in two Egyptian towns, but that required a team to pace out the long distance between them. Al-Bīrūnī wanted a measurement that one person could make from a single place.

He used a fact you can check yourself. From the top of a hill the horizon is not level with your eye. It lies a little below it. At the fort of Nandana he measured that small angle and found it was about 34 minutes of arc, a little over half a degree. He already knew the height of the hill.

That is enough, because your line of sight to the horizon only touches the Earth. Where it touches, it meets the radius at a right angle. So the hill, the horizon and the centre of the Earth form one right-angled triangle. If the hill has height h and the angle of dip is θ , then $\cos \theta = \frac{R}{R+h}$, and rearranging gives the radius:

$$R = \frac{h \cos \theta}{1 - \cos \theta}.$$

His answer was close to the modern value of 6371 km.

The same idea reaches things you can never travel to. Look at the Moon and it covers about half a degree of your view. That half degree is an angle at your eye. The Moon is about 384 000 km away, and multiplying that distance by the angle in radians gives the Moon's width: about 3500 km. The measured value is 3475 km.

One angle and one distance, and you have the size of a world. That is what this chapter builds. The triangle gives the sine and cosine rules. The solid gives volume and surface area. The circle gives the radian, the unit that links an arc to a full turn.

8.1 Three-Dimensional Geometry

Everything you know about coordinates in the plane extends to space by adding a third axis, z . The distance formula is the clearest example. In two dimensions it is Pythagoras' theorem; in three, it is Pythagoras applied *twice*, once across the base and once up.

KEY · IB The distance between (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

The midpoint is the average of the coordinates, $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$.

To see why, find the distance across the base first: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Treat that as one leg of a new right triangle, whose other leg is the height $(z_2 - z_1)$. A second use of Pythagoras gives the formula. A sphere is then just the set of points a fixed distance r from a centre (a, b, c) : $(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$.

The angle between a line and a plane

A line that is not parallel to a plane meets it at an angle, but which angle? The honest answer is the smallest one: the angle between the line and its own **shadow** on the plane, the projection you get by dropping perpendiculars from the line onto the surface.

DEF The **angle between a line and a plane** is the angle between the line and its perpendicular projection onto the plane. You find it by dropping a perpendicular from a point on the line to the plane, forming a right triangle, and using right-angled trigonometry.

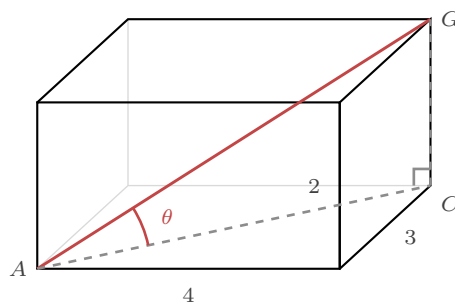


Figure 8.1 The space diagonal AG in scarlet, and its projection AC on the base, dashed. The vertical CG meets the base at a right angle, so ACG is a right triangle and θ is the angle between the diagonal and the base.

EXAMPLE 8.1

A cuboid has a 4×3 base and height 2. Find the angle between the space diagonal AG and the base.

The diagonal of the base is $AC = \sqrt{4^2 + 3^2} = 5$. This is the projection of AG onto the base, and $CG = 2$ is vertical, meeting the base at a right angle. In the right triangle ACG ,

$$\tan \theta = \frac{CG}{AC} = \frac{2}{5} \implies \theta = \arctan(0.4) \approx 21.8^\circ.$$

The key move is always the same: find the projection on the plane, then the perpendicular height, then use $\tan$.

The syllabus also asks for the angle between two intersecting *lines*, and in a solid the two lines are usually edges meeting at a vertex. Build the triangle they span, then solve it. The distance and midpoint formulas supply every length you need.

EXAMPLE 8.2

A square-based pyramid has base corners $A(0, 0, 0)$, $B(6, 0, 0)$, $C(6, 6, 0)$, $D(0, 6, 0)$ and apex $V(3, 3, 4)$. Find the angle between the slant edges VA and VB .

The two edges meet at V and span the triangle VAB , so the angle we want is $\angle AVB$. First measure the triangle. By the distance formula,

$$VA = \sqrt{(3-0)^2 + (3-0)^2 + (4-0)^2} = \sqrt{34},$$

and $VB = \sqrt{34}$ by the same calculation, while $AB = 6$ along the base.

The triangle is isosceles, so split it down the middle. The midpoint of AB is $M(3, 0, 0)$, and VM meets AB at right angles. In the right triangle VMA , the side $AM = 3$ is opposite the half-angle at V , and the hypotenuse is $VA = \sqrt{34}$:

$$\sin \frac{\theta}{2} = \frac{3}{\sqrt{34}} \implies \frac{\theta}{2} \approx 31.0^\circ \implies \theta \approx 61.9^\circ.$$

Everything came from the two formulas of this section: distances measured the triangle, the midpoint split it, and right-angled trigonometry finished it.

YOUR TURN 8.1 Three-Dimensional Geometry

1. Find the distance between each pair of points.

- | | | | |
|-----------------------------------|-----|----------------------------------|-----|
| (a) $(1, 2, 3)$ and $(4, 6, 3)$ | [2] | (b) $(0, 0, 0)$ and $(2, 3, 6)$ | [2] |
| (c) $(-1, 0, 2)$ and $(2, -4, 2)$ | [2] | (d) $(2, -1, 4)$ and $(5, 3, 4)$ | [2] |

2. Answer each part.

- (a) Find the midpoint of $(1, 2, 3)$ and $(5, 4, 1)$ [1]
 (b) Find the midpoint of $(2, -1, 4)$ and $(6, 3, 0)$. [2]

3. Write the equation of each sphere.

- (a) centre $(0, 0, 0)$, radius 5 [1] (b) centre $(1, 2, 3)$, radius 4 [2]
 (c) centre $(2, -1, 3)$, radius 6 [2] (d) centre $(0, 0, 2)$, radius $\sqrt{7}$ [2]

4. A cuboid has base 3×4 and height 12.

- (a) Find the length of the diagonal of the base. [2]
 (b) Find the length of the space diagonal. [2]
 (c) Find the angle between the space diagonal and the base. [3]

8.2 Volumes and Surface Areas of Solids

Beyond the cuboids, cylinders, and prisms you already know, three curved solids appear again and again. Their formulas are given in the exam. Knowing where each piece comes from is what prevents you from misusing them.

Sphere, radius r	$V = \frac{4}{3}\pi r^3, \quad A = 4\pi r^2$
Cone, radius r , height h	$V = \frac{1}{3}\pi r^2 h, \quad \text{curved } A = \pi r l$
Pyramid, base area B , height h	$V = \frac{1}{3}Bh$
Cylinder, radius r , height h	$V = \pi r^2 h, \quad \text{curved } A = 2\pi r h$

Two cautions save most errors. The cone's curved surface uses the **slant height** l , not the vertical height h ; the two are linked by Pythagoras, $l = \sqrt{r^2 + h^2}$. And a "total surface area" means every face. A solid hemisphere has a curved part $2\pi r^2$ and a flat circular base πr^2 , so its total is $3\pi r^2$. The same hemisphere as an open bowl has only $2\pi r^2$.

EXAMPLE 8.3

A solid is a cone of base radius 3 and height 4 sitting on top of a hemisphere of radius 3. Find its total surface area.

The exposed surface is the cone's curved part plus the hemisphere's curved part; the two flat circles where they join are hidden inside. The cone's slant height is $l = \sqrt{3^2 + 4^2} = 5$, so its curved

area is $\pi r l = 15\pi$. The hemisphere's curved area is $2\pi r^2 = 18\pi$. The total is

$$15\pi + 18\pi = 33\pi \approx 104.$$

Deciding which faces are exposed is the harder step. Once you know that, the formulas are routine.

YOUR TURN 8.2 Volumes and Surface Areas of Solids

5. Find the volume of each solid. Leave your answers in terms of π where appropriate.

- | | | | |
|---|-----|-------------------------------------|-----|
| (a) sphere of radius 3 | [1] | (b) cone of radius 3, height 4 | [2] |
| (c) pyramid with base area 20, height 9 | [1] | (d) cylinder of radius 4, height 10 | [2] |

6. Find the surface area asked for in each case.

- | | | | |
|--|-----|--|-----|
| (a) the surface area of a sphere of radius 5 | [1] | (b) the curved surface area of a cone of radius 5, slant height 13 | [1] |
| (c) the total surface area of a solid hemisphere of radius 4 | [3] | (d) the curved surface area of a cylinder of radius 4, height 10 | [2] |

7. A cone has radius 6 and height 8.

- | | |
|---|-----|
| (a) Find its slant height. | [1] |
| (b) Hence find its curved surface area. | [2] |

8. A solid is a cone of radius 3 and height 4 standing on a hemisphere of radius 3.

- | | |
|--|-----|
| (a) Explain which surfaces are exposed. | [2] |
| (b) Find the total surface area, in terms of π . | [3] |
| (c) Find the total volume, in terms of π . | [3] |

8.3 The Sine and Cosine Rules

Right-angled triangles are solved with $\sin$, $\cos$, and $\tan$: with sides labelled opposite, adjacent, and hypotenuse relative to an angle, $\sin \theta = \frac{o}{h}$, $\cos \theta = \frac{a}{h}$, $\tan \theta = \frac{o}{a}$. But most triangles have no right angle, and two rules extend trigonometry to all of them. Throughout, we label a side by the lower-case letter of the angle *opposite* it, so side a faces angle A .

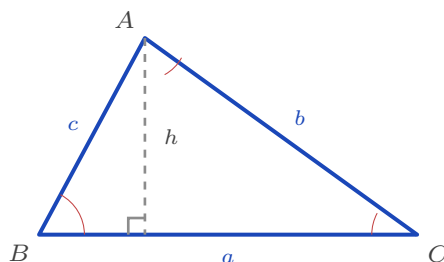


Figure 8.2 Each side is named after the angle opposite it: side a faces angle A , and so on. The dashed altitude h splits the triangle into two right triangles, which is where the sine rule comes from.

The sine rule

Drop the altitude h from A onto side a . It splits the triangle into two right triangles, and in each one h is the opposite side: $h = c \sin B$ and $h = b \sin C$. Setting these equal, $c \sin B = b \sin C$, which rearranges to $\frac{b}{\sin B} = \frac{c}{\sin C}$. Dropping a different altitude gives the third ratio.

KEY · IB **Sine rule.** In any triangle,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

The sine rule pairs a side with the angle facing it, so use it when you know such a pair, plus one more side or angle (cases AAS, ASA, or SSA).

The ambiguous case

The sine rule needs one extra check when you use it to find an *angle*.

Suppose you know two sides and an angle that is *not* between them. This is the SSA case. The sine rule then gives an equation of the form $\sin B = k$, and that equation has two solutions between 0° and 180° : the angle B , and its supplement $180^\circ - B$. A sine and its supplement are equal, so the equation cannot tell them apart.

Your calculator returns only the first one. You must test the second yourself. Add it to the angle you were given: if the total is still below 180° , a second triangle exists and both answers are correct.

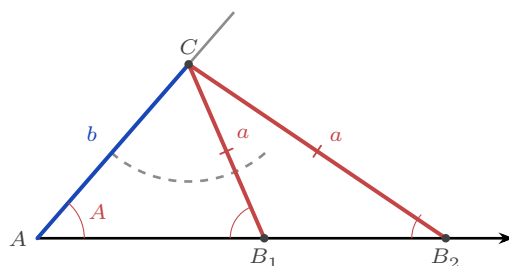


Figure 8.3 Two sides and a non-included angle. Swinging the fixed length a from C meets the base at B_1 and at B_2 , so two different triangles have the same a, b and A . The angles at B_1 and B_2 add to 180° , which is why the sine rule returns both.

EXAMPLE 8.4

In triangle ABC , $a = 8$, $b = 10$, and $A = 40^\circ$. Find angle B .

By the sine rule, $\frac{\sin B}{10} = \frac{\sin 40^\circ}{8}$, so

$$\sin B = \frac{10 \sin 40^\circ}{8} \approx 0.803.$$

The calculator gives $B \approx 53.5^\circ$. The supplement is $180^\circ - 53.5^\circ = 126.5^\circ$, so test it: $A + B = 40^\circ + 126.5^\circ = 166.5^\circ$, which is below 180° . A third angle of 13.5° remains, so that triangle exists too.

Both values of B are correct. There are two triangles, and a complete answer reports both.

CAUTION This check is needed only when the sine rule gives an angle. If you are finding a side, or using the cosine rule, no supplement arises. The cosine rule returns an angle between 0° and 180° directly, because $\cos$ is negative for obtuse angles and positive for acute ones, so it distinguishes the two.

The cosine rule

When you know two sides and the angle *between* them, no side-and-opposite-angle pair is available, and the sine rule cannot start. The cosine rule is used instead. It is Pythagoras' theorem with a correction term for the angle not being 90° . Coordinates prove it in three lines. Place the angle C at the origin with side a along the axis, so $B = (a, 0)$. Then A lies at $(b \cos C, b \sin C)$, by the definitions of cosine and sine. The distance formula measures the third side:

$$c^2 = (a - b \cos C)^2 + (b \sin C)^2 = a^2 - 2ab \cos C + b^2 \cos^2 C + b^2 \sin^2 C,$$

and since $\cos^2 C + \sin^2 C = 1$, the last two terms collapse to b^2 .

KEY · IB **Cosine rule.** In any triangle,

$$c^2 = a^2 + b^2 - 2ab \cos C, \quad \text{equivalently} \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

When $C = 90^\circ$, $\cos C = 0$ and the rule reduces to $c^2 = a^2 + b^2$, which is Pythagoras' theorem. The first form finds the third side from two sides and the included angle (SAS); the rearranged form finds an angle from all three sides (SSS).

The area of a triangle

The familiar area $\frac{1}{2} \text{ base} \times \text{height}$ becomes a trigonometric formula the moment you write the height in terms of a side. With base b and height $h = a \sin C$ (from the right triangle at C),

$$\text{Area} = \frac{1}{2} b (a \sin C).$$

KEY · IB **Area of a triangle.** Area = $\frac{1}{2} ab \sin C$, using two sides and the angle between them.

YOUR TURN 8.3 The Sine and Cosine Rules

9. Use the sine rule to find the required side or angle.

- | | |
|---|---|
| (a) $a = 7, A = 40^\circ, B = 60^\circ$: find b [2] | (b) $a = 9, A = 35^\circ, B = 55^\circ$: find b [2] |
| (c) $A = 45^\circ, B = 60^\circ, a = 10$: find C and b [3] | (d) $a = 10, b = 12, A = 35^\circ$: find the acute angle B [3] |

10. Use the cosine rule.

- | | |
|---|--|
| (a) $b = 5, c = 8, A = 60^\circ$: find a [2] | (b) $a = 6, b = 9, C = 110^\circ$: find c [3] |
| (c) $a = 6, b = 7, c = 8$: find angle A [3] | (d) sides 5, 6, 7: find the largest angle [3] |

11. Find the area of each triangle.

- | | |
|--|--------------------------------------|
| (a) sides 5 and 8, included angle 30° [2] | (b) $a = 9, b = 6, C = 90^\circ$ [1] |
| (c) sides 7 and 9, included angle 40° [2] | |

12. In triangle ABC , $a = 7, b = 9$ and $A = 45^\circ$.

- | | |
|---|-----|
| (a) Find the two possible values of angle B . | [3] |
| (b) Show that both give a valid triangle. | [3] |
| (c) Explain why this situation cannot arise when the given angle is included between the two given sides. | [2] |

8.4 Applications: Elevation, Depression, and Bearings

These rules are used most often in problems about the physical world, where the information is given as angles and directions rather than as a labelled triangle. Two conventions cover almost every case.

An **angle of elevation** is measured upward from the horizontal to a line of sight; an **angle of depression** is measured downward. The two horizontals are parallel, so the angle of depression from a cliff-top down to a boat equals the angle of elevation from the boat up to the cliff-top. They are alternate angles.

A **bearing** gives a direction as an angle measured clockwise from north. It runs from 000° to 360° , and it is always written with three figures, so due east is 090° and not 90° . The three figures make the reading unambiguous on a chart. Due south is 180° , due west is 270° , and north-west is 315° .

The **back bearing** is the direction of the return journey. It differs from the outward bearing by 180° . Add 180° if the bearing is less than 180° , and subtract 180° if it is 180° or more, so that the answer stays between 000° and 360° .

EXAMPLE 8.5

A ship sails 60 km from port P on a bearing of 050° to a point Q , then 80 km on a bearing of 140° to R . Find the distance PR .

The two bearings differ by $140^\circ - 50^\circ = 90^\circ$, so the ship turns through a right angle at Q : the angle PQR is 90° . Then PR is a hypotenuse, and Pythagoras gives

$$PR = \sqrt{60^2 + 80^2} = \sqrt{10000} = 100 \text{ km.}$$

Sketching the bearings from north lines at P and Q is what reveals the right angle; the arithmetic follows from the picture.

Most journeys do not turn so conveniently. When the angle at the corner is not 90° , the same sketch supplies the data for the cosine rule, and the sine rule then gives the direction back. Nearly every examination bearings question has this shape, so it is worth fixing the order of steps.

KEY Solving a bearings problem.

1. Sketch the journey, and draw a north line at **every** point where a bearing is given.
2. Mark each bearing as an angle clockwise from its own north line.
3. Find the interior angle of the triangle at the corner. Take the back bearing of the first leg, then subtract the second bearing.
4. Use the cosine rule for the missing distance, then the sine rule for the missing angle.
5. Turn that angle back into a bearing by adding it to the bearing of the first leg.

Step 1 is the one students leave out, and without it step 3 cannot be done.

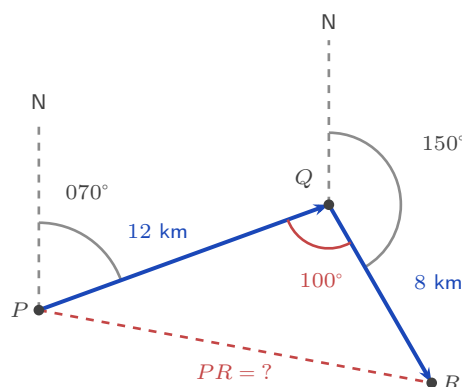


Figure 8.4 The two bearings are measured clockwise from a north line at P and at Q . Together they fix the interior angle $\angle PQR = 100^\circ$, which is what the cosine rule needs.

EXAMPLE 8.6

A hiker walks 12 km from P on a bearing of 070° to Q , then 8 km on a bearing of 150° to R . Find the distance PR and the bearing of R from P .

First construct the angle at the corner. The bearing back from Q to P is $070^\circ + 180^\circ = 250^\circ$, and the onward bearing is 150° , so the interior angle is $\angle PQR = 250^\circ - 150^\circ = 100^\circ$. Subtracting the back bearing is the standard method here, and the sketch confirms the result: the angle at the corner is clearly obtuse.

No right angle, so the cosine rule finds the closing side:

$$PR^2 = 12^2 + 8^2 - 2(12)(8) \cos 100^\circ \approx 241.3 \Rightarrow PR \approx 15.5 \text{ km.}$$

For the bearing of R from P , find the angle $\angle QPR$ by the sine rule:

$$\frac{\sin \angle QPR}{8} = \frac{\sin 100^\circ}{15.54} \Rightarrow \sin \angle QPR \approx 0.507 \Rightarrow \angle QPR \approx 30.5^\circ.$$

The bearing of R from P is therefore $070^\circ + 30.5^\circ \approx 100.5^\circ$, reported as 100° to the nearest degree. The labelled diagram carries the whole solution. The bearings give the corner angle, the cosine rule gives the closing side, and the sine rule gives the final direction.

YOUR TURN 8.4 Applications: Elevation, Depression, and Bearings

13. Find the angle of elevation in each case.

- (a) a 50 m tower, from 120 m away on level ground [2] (b) a 45 m tower, from 90 m away on level ground [2]

14. In each case, find the horizontal distance from the base of the tall object to the object below.

- (a) From the top of an 80 m cliff, the angle of depression to a boat is 25° . [2]
 (b) From the top of a 30 m building, the angle of depression to a car is 40° . [2]
 (c) From the top of a 60 m cliff, the angle of depression to a buoy is 20° . [3]

15. Answer each question about bearings.

- (a) Write the bearing of due west as a three-figure bearing. [1]
 (b) A ship travels on a bearing of 120° . Find its back bearing. [1]
 (c) B is on a bearing of 065° from A . Find the bearing of A from B . [2]
 (d) Find the back bearing of 200° , and explain why you subtract here rather than add. [2]

16. A ladder 6 m long rests against a vertical wall at 70° to the ground.

- (a) Find the height it reaches on the wall. [2]
 (b) Find the distance from the foot of the ladder to the wall. [2]

17. GDC A ship sails 40 km from P on a bearing of 030° to Q , then 50 km on a bearing of 120° to R .

- (a) Show that angle $PQR = 90^\circ$. [2]
 (b) Find the distance PR . [2]
 (c) Find the bearing of R from P . [3]

8.5 Radians, Arc Length, and Sector Area

The degree is a human invention: the Babylonians chose to cut the circle into 360 parts, and we inherited it. The circle itself suggests a more natural unit. Wrap the radius around the circumference; the angle it subtends at the centre is one **radian**. Since the circumference is $2\pi r$, exactly 2π radii fit around, so a full turn is 2π radians.

KEY 2π radians = 360° , so π radians = 180° . Convert by multiplying: degrees to radians $\times \frac{\pi}{180}$; radians to degrees $\times \frac{180}{\pi}$.

The advantage appears immediately, because arc length and sector area become simple fractions of the whole circle. An arc is the same fraction of the circumference as its angle θ is of a full turn: $\frac{l}{2\pi r} = \frac{\theta}{2\pi}$, and the 2π 's cancel.

KEY · IB For a sector of radius r and angle θ in radians,

$$\text{arc length } l = r\theta, \quad \text{sector area } A = \frac{1}{2}r^2\theta.$$

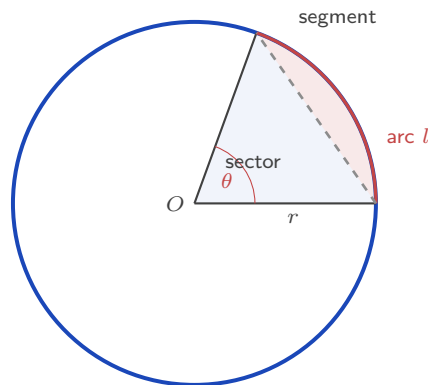


Figure 8.5 The shaded pie slice is the *sector*, bounded by two radii and the arc l . The smaller region cut off by the chord is the *segment*.

The area follows the same way: the sector is the fraction $\frac{\theta}{2\pi}$ of the whole disc πr^2 , giving $A = \frac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2}r^2\theta$. A **segment**, the region between a chord and its arc, is then a sector with the triangle cut away:

$$\text{segment} = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta = \frac{1}{2}r^2(\theta - \sin \theta).$$

EXAMPLE 8.7

A sector has radius 10 cm and angle $\frac{2\pi}{5}$ radians. Find its arc length and area.
Directly from the formulas,

$$l = r\theta = 10 \cdot \frac{2\pi}{5} = 4\pi \approx 12.6 \text{ cm}, \quad A = \frac{1}{2}r^2\theta = \frac{1}{2}(100)\frac{2\pi}{5} = 20\pi \approx 62.8 \text{ cm}^2.$$

For the segment cut off by the chord, subtract the triangle from the sector:

$$\frac{1}{2}r^2(\theta - \sin \theta) = 50 \left(\frac{2\pi}{5} - \sin \frac{2\pi}{5} \right) \approx 50(1.2566 - 0.9511) \approx 15.3 \text{ cm}^2.$$

All three formulas require θ in radians. Using degrees here is a common mistake, and it makes every answer wrong.

EXAMPLE 8.8

The opener mentioned the older method. Here it is in full. Eratosthenes measured the sun's rays at 7.2° to the vertical in Alexandria, while they were vertical in Syene, and took the arc between the towns to be 5000 stadia. Find the radius and circumference of the Earth in stadia.

Convert the angle first:

$$7.2^\circ \times \frac{\pi}{180} = \frac{\pi}{25} \text{ radians.}$$

The two towns and the Earth's centre make a sector whose arc is the distance between them, so $l = r\theta$ can be used in reverse:

$$5000 = r \cdot \frac{\pi}{25} \Rightarrow r = \frac{125\,000}{\pi} \approx 39\,800 \text{ stadia,}$$

and the circumference is $2\pi r = 250\,000$ stadia, which is just the one-fiftieth argument in formula form. Taking a stadion as roughly 158 m gives a circumference near 39 500 km, within a couple of percent of the true 40 075 km. One angle, one arc, one formula, and the planet is measured.

YOUR TURN 8.5 Radians, Arc Length, and Sector Area

18. Convert to radians, in terms of π .

- | | | | |
|-----------------|-----|-----------------|-----|
| (a) 60° | [1] | (b) 135° | [1] |
| (c) 210° | [2] | (d) 90° | [1] |

19. Convert to degrees.

- | | | | |
|---|-----|----------------------|-----|
| (a) $\frac{3\pi}{4}$ | [1] | (b) $\frac{2\pi}{3}$ | [1] |
| (c) 1 radian, to three significant figures[2] | | | |

20. Find the arc length and the area of each sector.

- | | | | |
|---------------------------------|-----|---|-----|
| (a) radius 8, angle 1.5 radians | [2] | (b) radius 6, angle $\frac{\pi}{3}$ radians | [2] |
| (c) radius 9, angle 2.4 radians | [3] | (d) radius 4, angle 90° | [3] |

21. In each case the angle is unknown. Find it from the information given.

- | | |
|--|-----|
| (a) A sector has radius 10 and arc length 25. Find its angle in radians. | [1] |
| (b) A sector has radius 6 and area 54. Find its angle in radians, then find its arc length.[3] | |

22. A chord subtends an angle of 2 radians at the centre of a circle of radius 8.

- (a) Find the area of the sector. [2]
- (b) Find the area of the triangle formed by the chord and the two radii. [2]
- (c) Find the area of the minor segment. [2]
- (d) Find the perimeter of the sector. [2]

Chapter 8 · Solids, Circles & Triangles

Reflections

IM Trigonometry was built by many people across many centuries. The Greek astronomer Hipparchus, around 150 BCE, made the first table of chords. Ptolemy extended it in the *Almagest*. In India, Aryabhata and those after him replaced the chord with the half-chord, which they called the *jyā*. That is our sine. The word travelled into Arabic as *jiba*, was mis-read as *jaib*, meaning “bay”, and was then translated into Latin as *sinus*. Islamic astronomers such as al-Bīrūnī developed the sine and cosine rules further, for their astronomy and for finding the direction of Mecca. The word “sine” still carries that chain of translations inside it.

TOK The formulas $l = r\theta$ and $A = \frac{1}{2}r^2\theta$ are only this simple when θ is in radians. Later you will meet another example: the derivative of $\sin x$ is exactly $\cos x$, again only in radians. The degree is a human choice. The Babylonians picked 360 because it divides so many ways. The radian is not a choice: it is the angle for which the arc equals the radius, and the circle itself fixes it. Is the radian discovered, then, where the degree is only invented?

RW Fixing an unknown position from measured angles is called **triangulation**, and it is still how we map the world. Surveyors once carried chains of triangles across whole countries. GPS finds your position from measured distances to satellites. Astronomers find the distance to nearby stars by *parallax*: as the Earth moves around the Sun, a near star seems to shift slightly against the far ones, and that tiny angle gives the distance. The unit it produces is called the parsec. Each of these is a triangle solved for a side nobody can reach.

EXT The ambiguous case is really a statement about congruence. Give a triangle by two angles and a side (ASA or AAS), or by three sides (SSS), or by two sides and the angle between them (SAS), and only one triangle fits. These are the congruence rules. But two sides and an angle that is *not* between them (SSA) is not a congruence rule. It can leave the triangle undecided, and the sine rule’s two answers are that gap showing through. The missing

congruence rule and the doubled answer are the same fact.

End-of-Chapter Problems — Chapter 8: Solids, Circles & Triangles

1. A cuboid has dimensions $6 \times 8 \times 10$.

- (a) Find the length of the space diagonal. [2]
- (b) Find the angle between the space diagonal and the base. [3]

2. Consider the points $A(2, 0, 0)$, $B(0, 4, 0)$ and $C(0, 0, 4)$.

- (a) Find the lengths AB , AC and BC . [3]
- (b) Find angle BAC . [3]

3. A cone has radius 5 and height 12.

- (a) Find the slant height. [1]
- (b) Find the volume. [2]
- (c) Find the total surface area. [2]

4. A solid consists of a cylinder of radius 3 and height 10 with a hemisphere of radius 3 on top. Find its total surface area. [5]

5. In triangle ABC , $a = 8$, $b = 11$, $c = 15$.

- (a) Find the largest angle. [3]
- (b) Find the area of the triangle. [2]

6. In triangle PQR , $p = 10$, $q = 7$ and $R = 95^\circ$.

- (a) Find r . [2]
- (b) Find the area of the triangle. [2]

7. A triangle has two sides of length 9 and 12 with an included angle of 40° .

- (a) Find the length of the third side. [2]
- (b) Find the area of the triangle. [2]

8. From port P , a ship sails 40 km on a bearing of 030° to Q , then 50 km on a bearing of 120° to R .

- (a) Show that angle $PQR = 90^\circ$. [2]
- (b) Find the distance PR . [2]
- (c) Find the bearing of R from P . [3]

9. A sector has radius 12 cm and angle 1.2 radians.

- (a) Find the arc length. [1]
- (b) Find the area of the sector. [2]
- (c) Find the perimeter of the sector. [2]

10. A sector has area 30 cm^2 and radius 6 cm.

- (a) Find the angle of the sector in radians. [2]
- (b) Find the arc length. [2]

11. A chord subtends an angle of 1.5 radians at the centre of a circle of radius 10 cm.

- (a) Find the area of the sector. [2]
- (b) Find the area of the triangle formed by the chord and the two radii. [2]
- (c) Find the area of the minor segment. [2]

12. In triangle ABC , $A = 50^\circ$, $a = 6$ and $b = 7$.

- (a) Find the two possible values of angle B . [3]
- (b) Find the two possible areas of the triangle. [3]

13. A regular hexagon is inscribed in a circle of radius 8. Find its area. [4]

14. A cone has volume 100π and radius 5.

- (a) Find the height. [2]
- (b) Find the slant height. [2]
- (c) Find the curved surface area. [1]

15. From a point on level ground, the angle of elevation of the top of a tower is 30° . Moving 40 m closer, the angle of elevation becomes 45° . Find the height of the tower. [5]

16. A square-based pyramid has base side 10 and height 12.

- (a) Find the volume. [2]
- (b) Find the slant height of a triangular face. [2]
- (c) Find the total surface area. [2]

17. The area of a triangle is 24, with $a = 8$ and $b = 10$. Find the two possible values of the included angle C . [3]

18. Two ships leave a port at the same time. One sails on a bearing of 040° at 12 km/h, the

other on a bearing of 100° at 16 km/h. Find the distance between them after 2 hours. [5]

19. Points $A(0, 0, 0)$ and $B(4, 4, 7)$ are given.

(a) Find $|AB|$. [2]

(b) Find the midpoint of AB . [1]

(c) Show that B lies on the sphere $x^2 + y^2 + z^2 = 81$. [1]

20. A pendulum of length 0.8 m swings through an angle of 0.5 radians. Find the length of the arc traced by the bob. [2]

21. Convert to radians, in terms of π : (a) 30° ; (b) 225° ; (c) 300° . [3]

22. Two observers stand 200 m apart on level ground, with a balloon in the vertical plane between them. From one the angle of elevation to the balloon is 40° ; from the other it is 55° . Find the height of the balloon. [6]

23. In triangle ABC , $AB = 12$, $BC = 9$ and angle $BAC = 35^\circ$. Find the two possible values of angle BCA . [4]

24. Convert to degrees: (a) $\frac{5\pi}{6}$; (b) $\frac{7\pi}{4}$; and find (c) the arc length of a sector of radius 9 and angle $\frac{5\pi}{6}$. [4]

25. A sector of a circle of radius 12 and angle 1.5 radians is rolled into a cone, so that the radius becomes the slant height and the arc becomes the base circumference. Find the radius and height of the cone. [6]

26. A sector of a circle has perimeter 20 and area 16. Find the radius r and angle θ . [6]

27. A chord of a circle of radius 10 has length 12. Find the area of the minor segment it cuts off. [6]

28. The area of a triangle is 84, and two of its sides have lengths 13 and 14. Find both possible lengths of the third side. [6]

29. GDC A yacht leaves harbour H and sails 25 km on a bearing of 048° to a buoy A . It then sails 18 km on a bearing of 130° to a second buoy B .

(a) Show that angle $HAB = 98^\circ$. [3]

(b) Find the distance HB . [4]

(c) Find the bearing of B from H . [4]

(d) Find the area of triangle HAB . [3]

- (e) The yacht returns directly from B to H . Find the total distance it has sailed. [2]

30. GDC A rectangular box $ABCDEFGH$ has a horizontal base $ABCD$ with $AB = 8$ cm and $BC = 6$ cm, and vertical edges of length 5 cm, so that G is the vertex above C .

- (a) Find the length of the base diagonal AC . [2]
 (b) Find the exact length of the space diagonal AG . [3]
 (c) Find the angle that AG makes with the base $ABCD$. [3]
 (d) Find the angle $\hat{GAB}$ between the diagonal AG and the edge AB . [4]
 (e) Find the total surface area of the box. [3]

31. GDC A container is a cone of base radius 6 cm and height 15 cm, held with its point downwards.

- (a) Find the volume of the cone when full, in terms of π . [2]
 (b) Water is poured in to a depth of 10 cm. Using similar triangles, find the radius of the circular water surface. [3]
 (c) Find the volume of water, in terms of π . [3]
 (d) Find the fraction of the cone's volume that is filled, and explain why it is not $\frac{2}{3}$ even though the water reaches $\frac{2}{3}$ of the height. [4]
 (e) Find the depth of water when the cone is exactly half full, giving your answer to three significant figures. [3]

Challenge Questions

32. How many triangles fit the data? In triangle ABC you are given the side b , the angle A , and the side a opposite A . This is the SSA case. This investigation finds exactly how many triangles fit, for every possible value of a .

Throughout, take A to be acute, and let h be the distance from C to the line AB .

- (a) Show that $h = b \sin A$. [2]
 (b) Take $b = 10$ and $A = 40^\circ$. Find the number of triangles when $a = 5$, when $a = 8$, and when $a = 12$. [4]
 (c) Explain why no triangle exists when $a < b \sin A$. (*Hint: what is the shortest distance from C to the line AB ?*) [3]
 (d) Explain why exactly one triangle exists when $a = b \sin A$, and state what is special about it. [3]
 (e) Show that two triangles exist exactly when $b \sin A < a < b$, and explain what goes wrong at $a = b$. [4]
 (f) Now let A be obtuse. State how many triangles exist for $a \leq b$ and for $a > b$, and justify each answer. [4]

33. The best cone from a sector. A sector of radius R and angle θ radians is cut from a sheet of paper. The two straight edges are joined to make a cone. This investigation finds the angle that gives the largest cone.

Write $t = \frac{\theta}{2\pi}$, so that $0 < t < 1$.

- (a) The arc of the sector becomes the circumference of the base. Show that the base radius is $r = Rt$. [3]
- (b) Show that the height of the cone is $h = R\sqrt{1-t^2}$. [3]
- (c) Hence show that the volume is $V = \frac{1}{3}\pi R^3 t^2 \sqrt{1-t^2}$. [3]
- (d) Taking $R = 1$, use technology to find the value of t that makes V largest, and hence find θ to the nearest degree. [4]
- (e) Show that at that maximum $\frac{r}{R} = \sqrt{\frac{2}{3}}$, and find the exact value of $\frac{h}{r}$. [5]

34. How reliable was al-Bīrūnī's measurement? The opener described his method.

From a hill of height h he measured the angle of dip θ to the horizon, and the radius of the Earth follows from one right-angled triangle. This investigation asks how much trust the answer deserves.

- (a) The line of sight touches the Earth, so it meets the radius there at a right angle. Show that $\cos \theta = \frac{R}{R+h}$, and hence that $R = \frac{h \cos \theta}{1 - \cos \theta}$. [3]
- (b) Al-Bīrūnī recorded $h = 652$ cubits and $\theta = 34'$. Find R in cubits. (*Hint: $34' = \frac{34}{60}$ of a degree*) [3]
- (c) Repeat the calculation for $\theta = 33'$ and for $\theta = 35'$. Find the percentage change in R in each case. [4]
- (d) For small θ the approximation $\cos \theta \approx 1 - \frac{1}{2}\theta^2$ holds, with θ in radians. Use it to show that $R \approx \frac{2h}{\theta^2}$. [3]
- (e) Hence explain why an error of about 3% in θ produces an error of about 6% in R . Comment on what this means for how confidently his result can be quoted. [4]

